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The Development of an Ultra-precision Dual-stage Based on a Master-Slave Control System

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***Abstract*— A ultra-precision dual-stage based on a master-slave control mode is presented to meet the demand for nanometer level accuracy in this paper. The dual-stage is constructed with a linear motor and a Lorentz motor. By adopting the air bearing, both linear motor and Lorentz motor are regarded as a double differential system respectively. The coupling system of two stages has been decoupled. The "PID + feedforward" control structure is adopted. Nanometer level positioning accuracy and high moving speed are achieved. The experimental results are provided.**

***Keywords-* Decouple; High speed/accuracy; Linear motor; Lorentz motor; Master-slave control; Ultra-precision dual-stage**

I. INTRODUCTION

Ever-increasing demands on higher productivity and better product quality in advanced industries, such as the semiconductors and precision engineering industries have continued to motivate and stimulate the development of high speed and high precision motion control system [1]. Since the late 1980s, a dual-stage system which is defined as a combination of coarse and fine stages has been introduced. A dual-stage system consists of a coarse stage driven by a coarse actuator and a fine stage driven by a fine actuator. The coarse actuator is of low bandwidth with large stroke, while the fine actuator is of high bandwidth with small stroke. Many research groups have proposed and manufactured dual actuator prototypes. Research efforts have also been devoted to the design of simple and high performance dual-actuator servo systems [2]-[5]. But there exists mechanical connection between its fine stage and coarse stage in those dual-stages, which brings problems to the fine stage due to uncertainties and disturbances transferred through its mechanical connection to the coarse stage. The positioning accuracy of the stage is just several decade nanometers.

A new master-slave motion control system based on air bearing is presented in this paper. By using two kinds of actuators —— a linear motor and a Lorentz motor, a new type dual-stage has been designed. Both rotors of two motors are guided and suspended respectively by air bearings which float on a granite surface for frictionless movement. Making full use of the advantages of the two actuators ——the long stroke of linear motor, the high accuracy of Lorentz motor, and the high response of both actuators, the stage completes positioning in one phase and assures both positioning accuracy and moving speed in long stroke. There is no friction force in both master and slave systems. But there exists reaction force between them. In other words those two systems compose a coupling system. By using decoupling technique, the couple system is decoupled two independent systems. The multiple input and multiple output (MIMO) system has been turned into two single input and single output (SISO) systems. Therefore the PID and other control methods can be applied. The "PID + feedforward" control structure is adopted in both stages. High position accuracy of nanometer degree and high moving velocity are achieved.

This paper begins by introduction in this section. Then the system configuration and motion equations of master/slave control in dual-stage are deduced in section 2. The decoupling of coupling master-slave system is in section3. The controller's structures of two kinds of motors are demonstrated in section 4. The results of the experiment are presented in section 5. Section 6 is the conclusion of this paper.

II. MODELING OF MASTER-SLAVE SYSTEM

As shown in Fig.1, the proposed stage is made up of a master system (fine stage) which is actuated by a Lorentz motor and a slave system (coarse stage) which is actuated by a linear motor. The rotors of two motors are all set on a great granite surface and guided by air-bearings respectively. The stator of the Lorentz motor (coil module) is attached to the rotor of the linear motor. A reflector of laser interferometer is set in the rotor (magnet module) of the Lorentz motor. The fine stage uses the laser interferometer as its position test device. Hall element and magnet are set in the rotor of the linear motor and the rotor of the Lorentz motor respectively, thus forming a sensor named difference sensor that is used to measure the distance of linear motor relative to Lorentz motor.

The master-slave control system composed of linear motor and Lorentz motor is a macro-micro motion control system. Its procedure of motion control is as follows: First the set point of master (micro) system composed of Lorentz motor and the zero position (relative to master system) of slave (macro) system composed of linear motor are given. When the system starts motion, master control system uses the laser interferometer as its position measurement equipment to motion according to motion trajectory beforehand. After a sampling period, the slave system uses

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difference sensor doing position tracking control in order to keep the zero position to master system while the master system still does deviation control according to named motion trajectory. As the motion carries out, the deviation of actual position to theory position is becoming smaller and smaller. When the deviation is zero, it denotes that the named theory position is reached. So the result motion of master-slave is like to amplify the stroke of the master (micro) system composed of Lorentz.

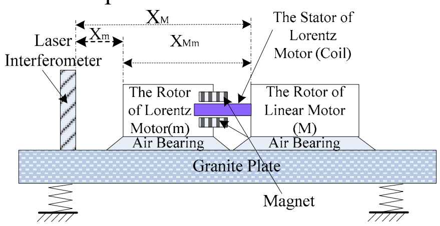


Figure 1. Experiment configuration

In Fig.1 granite surface and air bearing are capable of sustaining a high load and make the dual-stage frictionless. There is no mechanical connection in the two stages. In vertical direction the supporting forces from air bearing equal to the gravity of each stage. We can simplify Fig.1 by taking no account of forces in vertical direction. Fig.2 shows the horizontal forces of dual-stage. The system consists of two physical unconnected masses. One is m control system, i.e. Lorentz motor control system. The position of m relative to granite plate X_m is measured by a laser interferometer. The desired position of m is obtained by controlling the force f as a function of the deviation of Lorentz motor position. This control system is regarded as the main control system or master system. The other system is slave control system, i.e. linear motor control system. Its desired position is obtained by controlling force F that is the function of the deviation of position X_M (X_m-X_{Mm}). The purpose of the control is to keep M in track of m.

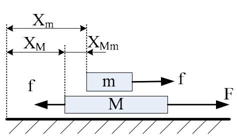


Figure 2. Model of master-slave control

Owing to using air bearing both the linear motor and the Lorentz motor are regarded as a double differential system. Their motion equations are

$$\begin{cases} F - f = M\ddot{x}_M \\ f = m\ddot{x}_m \end{cases} \qquad (1)$$

where, M is the total mass of linear motor's rotor and the stator of Lorentz motor. m is the total mass of Lorentz motor's rotor and the laser interferometer's reflector. F and f denote the driving force of linear motor and Lorentz motor respectively. X_m denotes the position of Lorentz motor relative to granite foundation measured directly by interferometer. X_{Mm} is the position of linear motor relative to Lorentz motor measured by a difference sensor. X_M is the position of linear motor relative to granite foundation measured indirectly by interferometer and difference sensor.

From Eq. (1), the Laplace motion equation can be obtained:

$$\begin{cases} X_m = \dfrac{1}{ms^2} f \\ X_{Mm} = X_M - X_m = \dfrac{1}{Ms^2} F - \dfrac{(M+m)}{Mms^2} f \end{cases} \qquad (2)$$

It is obvious that the system represented by Eq. (2) is a coupling system. It is necessary to decouple it first and then the SISIO control system can be designed.

III. DCOUPLING OF THE MASTER-SLAVE SYSTEM

The state and output equations of a MIMO linear time-invariant system with output-feedback have the form (see Fig.3)

$$\dot{x}(t) = Ax(t) + Bu(t) - Hy(t) \qquad (3)$$

$$y(t) = Cx(t) \qquad (4)$$

where $x \in R^n$ is the state vector, $u \in R^m$ is the input vector, $y \in R^m$ is the output vector, $A \in R^{n\times n}$, $B \in R^{n\times m}$, $C \in R^{m\times n}$.

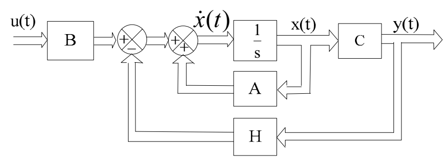


Figure 3. Block diagram of state space model with output-feedback

The closed-loop state equation obtained by combining Eq. (3) and Eq. (4) yields

$$\dot{x}(t) = (A - HC)x(t) + Bu(t) \qquad (5)$$

It is obvious that the system represented by Eq. (3) and Eq.(4) is a coupling system. It is necessary to decouple it first and then the SISIO control system can be designed.

Fig.3 can be transformed into Fig.4 which is denoted in transfer function form because SISO system is usually denoted in that form.

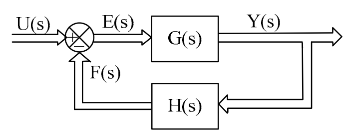


Figure 4. Block diagram of transfer function with output-feedback

From Fig.4 we can get

$$Y(s) = G(s)(U(s) - F(s)) = G(s)(U(s) - H(s)Y(s))$$

So the transfer matrix of the closed-loop system is

$$\Phi(s) = \frac{Y(s)}{U(s)} = \left[I + G(s)H(s)\right]^{-1} G(s) \qquad (6)$$

The series compensator method was adopted to decouple the master-slave system. The method decouples the coupling system by connecting in series a compensator to forward passage. Fig.5 shows the block diagram of series compensator method.

In Fig.3, $G_p(s)$ is the transfer matrix of coupling system. $G_c(s)$ is the matrix of series compensator. H（s）is feedback matrix. E(s) is the error vector.

The forward transfer matrix is

$$G(s) = C[sI - A]^{-1} B = G_P(s) G_C(s) \tag{7}$$

The matrix of series compensator can be expressed as

$$G_C(s) = G_P^{-1}(s)\Phi(s)[I - H(s)\Phi(s)]^{-1} \tag{8}$$

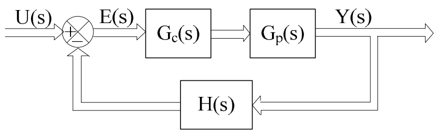


Figure 5. Block diagram of series compensator decouple

where, Φ(s) is the transfer matrix of closed-loop.

For unit feedback system, H（s）=1, So

$$\Phi(s) = [I + G(s)]^{-1} G(s) \tag{9}$$

$$G_C(s) = G_P^{-1}(s)\Phi(s)[I - \Phi(s)]^{-1} \tag{10}$$

From Eq.(12) the transfer matrix of unit feedback decouple system can be solved

$$G(s) = G_c(s) G_P(s) = \Phi(s)[I - \Phi(s)]^{-1} \tag{11}$$

As for coupling master-slave system, the transfer matrix of master-slave couple system ($G_p(s)$) is

$$G_P(s) = \begin{bmatrix} \frac{1}{ms^2} & 0 \\ -\frac{m+M}{mMs^2} & \frac{1}{Ms^2} \end{bmatrix} \tag{12}$$

The transfer matrix of close loop (Φ(s)) is

$$\Phi(s) = \begin{bmatrix} \frac{1}{ms^2+1} & 0 \\ 0 & \frac{1}{Ms^2+1} \end{bmatrix} \tag{13}$$

The matrix of series compensator can be gotten by taking Eq.(12) and Eq.(13) into Eq.(10):

$$G_C(s) = \begin{bmatrix} 1 & 0 \\ \frac{m+M}{m} & 1 \end{bmatrix} \tag{14}$$

Fig. 6 is the decouple block diagram of master-slave control system. From Fig.6 we can know that the coupling master-slave system has been turned from a MIMO (two-inputs and two-outputs) into two SISO systems by decoupling. Thereby both the master and slave control systems can be designed according to SISO control system.

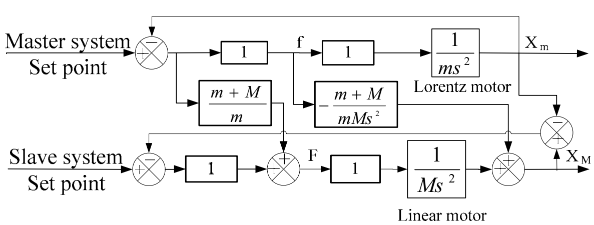


Figure 6. Decouple block diagram of master-slave control system Block

IV. THE CONTROLLER STRUCTURE

A. Linear Motor and Lorentz Motor

The linear motor used in this paper is permanent magnet linear synchronous motor (PMLSM) with air bearing.

Lorentz motor is the same as the linear motor in work principle. It consists of magnet module and coil module. It also makes use of Lorentz principle. But it has a relative short stroke (0.5-2mm) with very fast response and very high positioning accuracy in its whole stroke. Fig.7 illustrates the mechanical response curve (Bode plot) of the Lorentz motor (with the air bearing) in this paper. It is obviously a double differential plant.

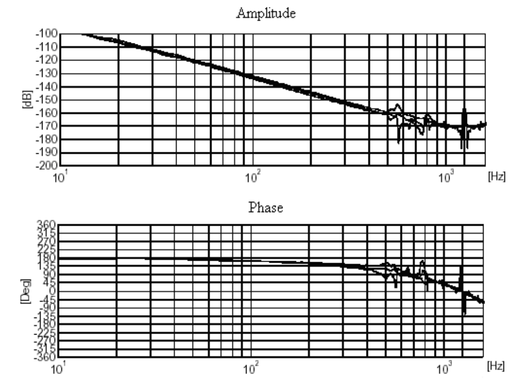


Figure 7. The mechanical transfer function of Lorentz motor

B. The Controller of two Motors

Since the linear motor and Lorentz motor used in this paper all have air bearing, both of the two motors are regarded as a double differential plant [1]. They use similar controllers expressed as "PID + low pass filter + notch filter + Feedforward".

A second-order low pass filter is used to eliminate the disturbance of high frequency noise. Acceleration feedforward is used to improve the tracing accuracy and to shorten the adjusting time. To eliminate the disturbance in some specific frequency, notch filters are used. Fig.8 shows the block diagram of one actuator (linear motor or Lorentz motor) controller.

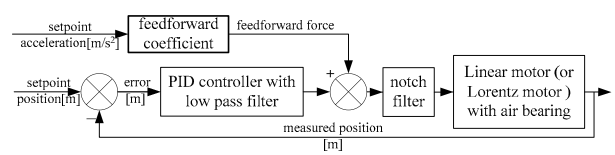


Figure 8. Block diagram of one actuator controller

Since the structure of the forward passage of controller is "PID + second-order low pass filter + notch filter" the transfer function of the controller is

$$G_C(s) = \frac{K_p\left(\frac{s^2}{(2\pi f_D)} + s + (2\pi f_I)\right)}{\left(\frac{s}{2\pi f_{lwp}}\right)^2 + 2\zeta_{lwp}\frac{s}{(2\pi f_{lwp})} + 1} \tag{15}$$

where numerator is PID controller. Denominator is second-order low pass filter. K_P, $K_P*2\pi f_I$, and $K_P/(2\pi f_D)$ are proportional, integrator and differentiator coefficient. f_{lwp} and ζ_{lwp} are the low pass frequency and the damp ratio of the second-order low pass filter respectively.

Transfer function of notch filter is

$$H(s)=\frac{\omega_{pole}^2}{\omega_{zero}^2}\frac{s^2+2\zeta_{zero}\omega_{zero}s+\omega_{zero}^2}{s^2+2\zeta_{pole}s+\omega_{pole}^2} \tag{16}$$

C. *The parameters of linear Motor and Lorentz Motor*

The parameters of master control system and slave control system are listed in Table I and Table II.

TABLE I. PARAMETERS OF MASTER CONTROL SYSTEM

Kp	[kN/m]	482.000
Integrator Frequency	[Hz]	4.000
Differentiator Frequency	[Hz]	40.000
Low Pass Frequency	[Hz]	300.000
Low Pass Dump	[Hz]	0.741
Motor Constant	[N/A]	28.000
Max Velocity	[m/s]	0.200
Max Acceleration	[m/s^2]	6.000
Jerk	[m/s^3]	200.000
Setpoint Feedforward	[kg]	18.300
Bandwidth	[Hz]	160.000

TABLE II. PARAMETERS OF SLAVE CONTROL SYSTEM

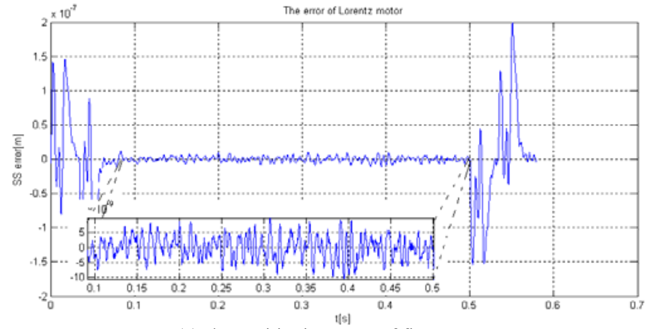
Kp	[kN/m]	47.600
Integrator Frequency	[Hz]	0.500
Differentiator Frequency	[Hz]	7.000
Low Pass Frequency	[Hz]	40.000
Low Pass Dump	[Hz]	0.741
Motor Constant	[N/A]	55.000
Bandwidth	[Hz]	19.000

V. EXPERIMENTS

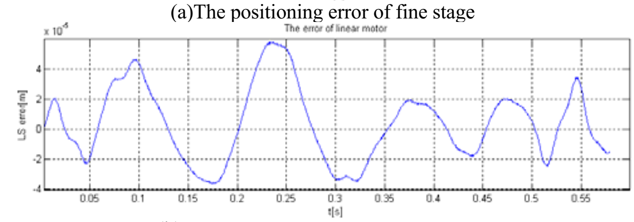
We have carried out an actual experiment in the dual-stage consisting of the linear motor and the Lorentz motor. The measuring accuracy of the laser interferometer is 0.6 nanometer, while the measuring accuracy of the difference sensor is 0.001 millimeter. The delay correction time of feedforward is 360 microseconds.

The power amplifiers provide a three phase output current proportional to the input voltage. The phase current range of linear motor is ± 14 ampere peak. The phase current range of Lorentz motor is ± 8.5 ampere.

Fig.9 shows the positioning error of fine stage (master control system) and coarse stage (slave control system). We can know that the positioning error of fine stage at stable speed is rather small. Its peak value is not larger than 10 nm and its adjust time is also short. The positioning error will be stable at nm level just after about 10ms while the positioning error of coarse stage is relatively large and its max value reaches 0.0486mm. Since the value is in the motion stroke of Lorentz motor and the positioning error of dual stage is determined by the positioning accuracy of fine stage, it will not affect the positioning accuracy of whole control system.



(a)The positioning error of fine stage



(b) The positioning error of coarse stage

Figure 9. The positioning error of dual-stage

VI. CONCLUSIONS

In this paper we present a dual-stage based on a master-slave control system. By using air bearing, there was no mechanical connection between two stages. The motion equations of the dual-stage were deduced. The series compensator method was adopted to decouple the master-slave system. The nanometer level positioning accuracy, high speed and long stroke of dual-stage were achieved.

The control methods discussed in this paper have been successful in applying in the national “863” key project of China —‘100nm step & scan projection lithography’.

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